

High-Temperature Impact-Ionization Model for 4H-SiC

Selamnesh Nida^{ID} and Ulrike Grossner^{ID}, *Member, IEEE*

Abstract—Silicon carbide (4H-SiC) devices experiencing avalanche conditions can reach temperatures above 1500 K. Simulation of impact ionization in devices should, therefore, include models valid up to such high temperatures. However, calibrations of impact ionization coefficients are available only up to 580 K, and simulations of switching show deviations from measurements at higher temperatures. In this paper, a more accurate model based on the underlying physics of high temperature and anisotropic avalanche generation is proposed. This model enables calibrations performed at room temperature along the *c*-axis to be more precisely extrapolated to any temperature of interest in 2-D device simulations. The model provides an excellent fit to the measurements of breakdown voltages during unclamped inductive switching tests of power MOSFETs, enabling a more accurate prediction of ruggedness. The more comprehensive physics included in the model makes it also applicable to other wide bandgap semiconductors such as GaN, diamond, and Ga₂O₃, which also exhibit a high critical electric field.

Index Terms—Avalanche, electrothermal simulation, impact ionization, modeling, reliability, ruggedness.

I. INTRODUCTION

SILICON carbide (SiC) is a wide bandgap material, making it well suited for high-voltage device applications. The temperature where the blocking capability is lost, the intrinsic temperature, T_{int} , is much higher in 4H-SiC, ≈ 1500 K [1], than in silicon, ≈ 450 K [2]–[5].

One of the failure modes against which devices need to be rugged is avalanche multiplication. In the presence of high electric field, the impact ionization of valence electrons generates electron–hole pairs. Devices experience such high electric fields during turning-off inductive loads where the back electromotive force (EMF) drives the switching device into avalanche. Thus, device designers need models of impact ionization accurate up to T_{int} to correctly consider electrothermal ruggedness in their designs. Accurate determination of the ionization coefficient is important to estimate the breakdown voltage of devices and the change of the breakdown voltage with temperature during transient losses.

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The authors are with the Advanced Power Semiconductor Laboratory, Department of Information Technology and Electrical Engineering, ETH Zurich, 8092 Zurich, Switzerland (e-mail: nida@aps.ee.ethz.ch; ulrike.grossner@ethz.ch).

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The state-of-the-art impact ionization models are originally proposed and calibrated for silicon and narrow bandgap materials [6]–[9]. These models are adapted for 4H-SiC and parameters are extracted from photoionization experiments [10]–[13]. However, parameters for 4H-SiC are calibrated only up to 580 K despite devices reaching above 1500 K during dissipative transients [1].

In addition, unlike silicon, 4H-SiC shows anisotropic transport properties due to its crystallographic structure [14]. The anisotropy of impact ionization in 4H-SiC was demonstrated and models were calibrated with separate photomultiplication experiments along the *c*-axis $\langle 0001 \rangle$ and the *a*-axis $\langle 11\bar{2}0 \rangle$ by Hatakeyama [10].

In this paper, we propose a model for the impact ionization coefficient which incorporates the temperature dependence and anisotropy of impact ionization in 4H-SiC. This model is based on previous works of Keldysh [6] and Thornber [15] on the theory of impact ionization, which were originally proposed for silicon. The influence of the high critical electric field of 4H-SiC is incorporated for the first time in a 4H-SiC impact ionization model. The effect of temperature as well as anisotropy of 4H-SiC are proposed in a novel way and show excellent fit to experimental data obtained using unclamped inductive switching (UIS) tests.

The main advantage of the model is that calibration along one crystal axis at room temperature is sufficient to give the most accurate impact ionization coefficients available up to T_{int} . Therefore, the proposed model can be used for 2-D-device analysis.

This paper is organized as follows. In Section II, the state-of-the-art avalanche models used for 4H-SiC as well as the one proposed in this paper are presented. The incorporation of temperature and anisotropy in the proposed model is described. In Section III, the impact ionization coefficient along $\langle 0001 \rangle$ at elevated temperatures from the proposed model is compared with those given in the previous literature [10]–[13], [16]. In Section IV, the measurement of a power MOSFET drain–source voltage under avalanche breakdown conditions is used to verify the proposed model and compare with the state of the art.

II. PROPOSED IMPACT IONIZATION MODEL

The impact ionization process is included in the drift diffusion transport as a carrier generation rate G_{av}

$$G_{\text{av}} = \frac{1}{q}(\alpha_n J_n + \alpha_p J_p) \quad (1)$$

where $J_{n,p}$ is the electron-hole current density and $\alpha_{n,p}$ is a temperature-dependent ionization coefficient for electrons and holes, respectively. It represents the average number of ionizing collisions per unit distance and is proportional to the electric field. Thus, the generation rate is calculated using the electric field in the direction of the current. The state-of-the-art avalanche models describe this rate, α . A phenomenological model from Chynoweth [17] modified by Okuto and Crowell [18], is given as follows:

$$\alpha = aF^\rho \exp\left(\left(\frac{-b}{F}\right)^\delta\right) \quad (2)$$

where F is the electric field and the coefficients a and b are temperature-dependent parameters. The state-of-the-art impact ionization models in 4H-SiC [10]–[13] use this formalism, where the coefficients are calibrated with photoionization experiments.

A comprehensive model which takes into account more physical parameters was proposed by Thornber [15]. α in Thornber's model is determined as

$$\alpha = \frac{qF}{\epsilon_{ii}} \exp\left(\frac{-\epsilon_{ii}}{qF\lambda + \frac{q^2 F^2 \lambda^2}{3\langle\epsilon_r\rangle} + kT}\right) \quad (3)$$

where F is the electric field and ϵ_{ii} is the threshold energy for impact ionization [15]. This model was previously used by Konstantinov *et al.* [16]. The mean free path of the nonionizing scattering process is given by λ . The prefactor, qF/ϵ_{ii} , describes the ionization coefficient in the absence of nonionizing scattering mechanisms. The exponential function describes the probability of ionizing scattering events. λ and ϵ_{ii} are temperature dependent.

In the case electrons in 4H-SiC, at avalanche conditions, the major scattering mechanism is that of scattering with the longitudinal optical (LO) phonon [14]. From Raman experiments, this energy, ϵ_r , is determined to be 120 meV [19]. $\langle\epsilon_r\rangle$ represents the average energy lost per scattering. In the case of holes in 4H-SiC, results have indicated that the optical deformation potential scattering dominates with an energy of 85 meV [20]. Due to the complexity of the semiconductor band structure and the corresponding ionizing transitions, λ of these scattering mechanisms and ϵ_{ii} at 300 K are used as fitting parameters.

A. Incorporation of Anisotropy

The 4H-SiC shows anisotropic transport properties such as mobility, velocity saturation, as well as impact ionization [14]. In order to perform simulations of real devices, the anisotropy of impact ionization needs to be considered. Typically, 4H-SiC wafers are grown on $\langle 0001 \rangle$ substrates with a 4° off angle. Furthermore, since the $\langle 11\bar{2}0 \rangle$ direction is a natural cleavage plane of 4H-SiC, a lower defect density at the semiconductor/oxide interface can be expected. Thus, these two orientations are the orientations of the main interest in 4H-SiC for 2-D simulations.

The presence of anisotropy in impact ionization is included in the formalism by Keldysh [6] where the anisotropic electric

field, F , is replaced by an effective isotropic electric field, $F^* = (m/m_{||})^{1/2} F$. In this case, m is the effective mass density of states and $m_{||}$ is the effective mass along the direction of the force F . Thus, the anisotropy can be included in Thornber's formulation, (3), by substituting F^* in place of F .

This formalism by Keldysh, however, only considers anisotropy of effective mass, which is measured at the low applied field. However, in 4H-SiC band, edge phonons along the c -axis scatter with energetic electrons-holes moving along this axis which gives a different scattering rate compared with the a -axis perpendicular to it. The anisotropy of impact ionization, therefore, is different at higher applied fields. This is evident where the electron velocity saturation is larger in the $\langle 11\bar{2}0 \rangle$ direction, $V_{\text{sat}\langle 11\bar{2}0 \rangle} / V_{\text{sat}\langle 0001 \rangle} = 1.42$ [21], [22], while the ratio of inverse effective mass is smaller, $(m_{\langle 11\bar{2}0 \rangle}^* / m_{\langle 0001 \rangle}^*)^{-1/2} = 0.91$. Since the anisotropy of the impact ionization is the result of both effective mass and scattering rates at high electric field, using anisotropy of velocity saturation would capture the phenomenon more accurately.

Therefore, the proposed model determines the effective isotropic electric field, F^* from anisotropy of velocity saturation. F^* is considered to be proportional to V_{sat} in the corresponding crystal orientation.

The anisotropy of the prefactor, qF/ϵ_{ii} , which refers to impact ionization coefficient in the absence of scattering, is modeled with the anisotropy of effective mass.

The coefficient, α , is adapted to high temperature and different crystallographic orientations based on the underlying physics, as described in the following.

B. Temperature Dependence

The temperature dependence of α in Thornber's model derives from the kT term as well as the change of λ , $\langle\epsilon_r\rangle$, and ϵ_{ii} as a function of temperature. As temperature rises, the number of LO phonons and collision of carriers with them increase, and this decreases the mean free path, λ , of the scattering. The change in λ at temperature T is

$$\lambda(T) \propto \tau \propto (2N + 1)^{-1} \propto \tanh(\epsilon_r/2kT). \quad (4)$$

However, this derivation originated from Fermi's Golden Rule which is only valid as long as it does not violate Heisenberg's uncertainty principle. In the case of holes in 4H-SiC, the high scattering rate results in collision broadening [23]–[25]. To lessen the numerical burden of incorporating collision broadening, a phenomenological model is given in Bude's chapter in [25] in which the scattering rate, (τ_{highrate}) is changed to $(\tau_{\text{highrate}}^{1/2})$ [20], [25]. Therefore, λ becomes

$$\lambda(T) = \lambda_{0K} \sqrt{\tanh(\epsilon_r/2kT)}. \quad (5)$$

To the best of author knowledge, the above-mentioned effect of collision broadening on impact ionization coefficient of 4H-SiC is included in avalanche models for the first time.

The rise in temperature also reduces $\langle\epsilon_r\rangle$, the average energy lost to LO phonon collisions. At higher temperatures,

TABLE I
FORMULAS IN THE PROPOSED MODEL

$$\alpha = \frac{qF}{\epsilon_{ii}} \exp\left(\frac{-\epsilon_{ii}}{qF\lambda + \frac{q^2 F^2 \lambda^2}{3(\epsilon_r)} + kT}\right)$$

$$\lambda(T) = \lambda_{300K} \sqrt{\tanh(\epsilon_r/2kT) / \sqrt{\tanh(\epsilon_r/2k(300K))}}$$

$$\epsilon_{ii}(T) = \epsilon_{ii}(300K) E_g(T) / E_g(300K)$$

the probability of an electron absorbing a phonon increases and the average energy lost, $\langle\epsilon_r\rangle$, decreases analogous to λ . The ratio, $\lambda/\langle\epsilon_r\rangle$, remains constant [26].

The temperature coefficient of ϵ_r , the LO phonon energy (120 meV at 300 K), is negligible in 4H-SiC. From Raman experiments by Peng *et al.* [27], a $\Delta\epsilon_r \approx 0.13$ –0.2-meV shift is measured between 240...660 K for samples with doping concentration of 2.42×10^{18} – 1.84×10^{17} cm⁻³, respectively. Extrapolation of the change to 1500 K and incorporating it did not show a change in the calculated V_{BR} values in this paper. Therefore, in this paper, the value of ϵ_r is assumed constant. The energy of optical deformation potential is also assumed constant in the proposed model. More accurate measurements at higher temperatures are currently not available.

On the other hand, with increased temperature, the bandgap of 4H-SiC reduces. With the threshold energy assumed to be proportional to the bandgap [28], the threshold energy decreases accordingly. This is incorporated into the proposed model in the following equation with $E_g(T)$ parameters in 4H-SiC taken from [29]:

$$\epsilon_{ii}(T) \propto E_g(T). \quad (6)$$

A complete temperature dependence of α is then extrapolated from coefficients of (3) at 300 K, (5) and (6) (see Table I). The anisotropy of impact ionization is assumed to remain constant with temperature. Previous literature has shown no temperature dependence of mobility anisotropy in 4H-SiC samples up to 600 K [30]. However, the temperature dependence of anisotropy up to T_{int} of 4H-SiC would have to be investigated in detail in the future.

III. COMPARISON WITH THE STATE OF THE ART

State-of-the-art experiments investigating impact ionization in 4H-SiC have been performed for the past two decades. The parameters for α are calibrated through photomultiplication experiments. The first experiment was performed by Konstantinov *et al.* [16] Raghunathan and Baliga [11] performed electron beam induced current-based experiments to extract parameters for holes more precisely.

The experiments were further improved to get pure hole multiplication by removing excess noise from electrons by Loh *et al.* [13]. Niwa *et al.* [12] performed multiplication experiments on p⁺pn⁺ and n⁺np⁺ diodes. The anisotropy of impact ionization was experimentally demonstrated by Hatakeyama [10] with independent photomultiplication experiments along both (0001) and (11 $\bar{2}$ 0).

Results from the photomultiplication experiments given in the literature [10]–[13], [16] were fitted using the

TABLE II
PARAMETERS RANGES FOR THE PROPOSED MODEL 300 K

Parameter	λ_n (cm)	λ_p (cm)	ϵ_{ii_n} (eV)	ϵ_{ii_p} (eV)
Range	$(1.685\text{--}2.7) \times 10^{-7}$	$(3.16\text{--}4.41) \times 10^{-7}$	4.998–10.61	6.357–16.17
Value used	2.7×10^{-7}	3.949×10^{-7}	10.61	10.87

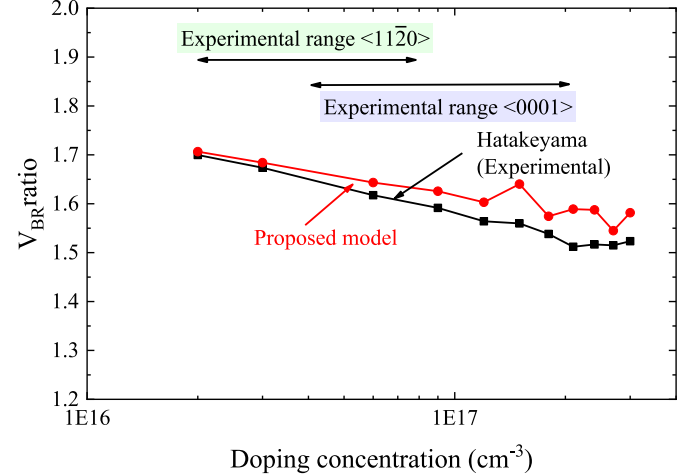


Fig. 1. Anisotropy of impact ionization, demonstrated by the $V_{BR\langle 0001 \rangle} / V_{BR\langle 11\bar{2}0 \rangle}$ of p-i-n diodes, in the proposed model compared with results from Hatakeyama [10]. Experimental ranges of Hatakeyama's result are shown by shaded rectangles. Part of the experimental error may be due to nonoverlapping ranges.

proposed formula at 300 K for 4H-SiC along (0001). Since the avalanche in 4H-SiC is a hole dominated process, there is more uniformity in α_p from various experiments compared to electron multiplication, α_n . Experimental conditions of these results have been discussed in the respective publications. In Table II, the extracted range of the parameters at 300 K as well as the parameter set that fits the measurements of breakdown voltage in this paper are given.

A. Anisotropy of Breakdown Voltage

The anisotropy of 4H-SiC impact ionization determined from the proposed model is compared with the results demonstrated by Hatakeyama [10]. The breakdown voltage of p-i-n diodes with varying doping concentrations on a (11 $\bar{2}$ 0) and (0001) substrate was measured by Hatakeyama [10]. Accordingly, simulation results using Hatakeyama's model and the proposed model, described in Section II-A, are presented in Fig. 1. The range of doping used in Hatakeyama's experiments is shown in the shaded regions of Fig. 1. The proposed model gives promising agreement with experimental results in capturing anisotropy of impact ionization. Therefore, it is used further for extrapolation of impact ionization coefficient at high temperatures.

B. Coefficients at Elevated Temperatures

Photomultiplication experiments in 4H-SiC at elevated temperatures are performed by Hatakeyama [10], Raghunathan and Baliga [11], and Niwa *et al.* [12] along the

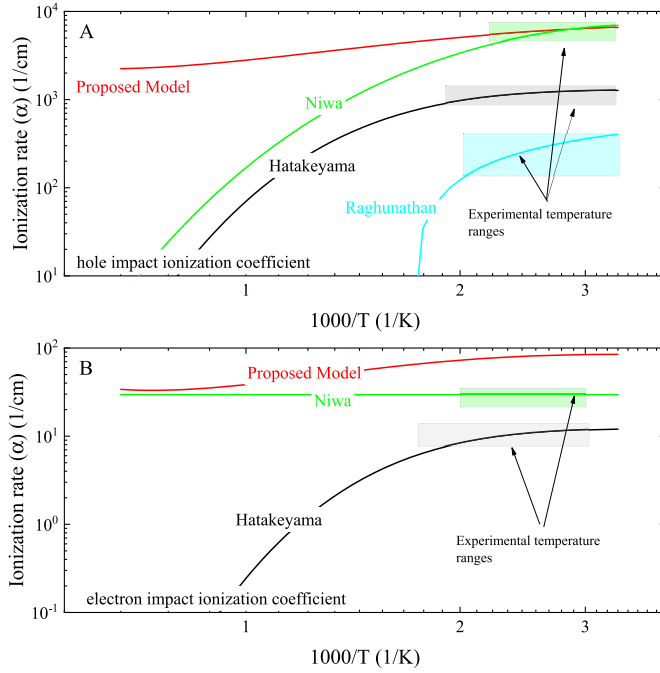


Fig. 2. Comparison of impact ionization coefficients along (0001) in 4H-SiC as a function of temperature from different models at $F = 2 \times 10^6$ V/cm, for (a) holes and (b) electrons.

(0001) direction. These experiments are performed only up to 580 K. Therefore, in the state-of-the-art failure analysis, the value of $\alpha_{n,p}$ is extrapolated up to T_{int} from these calibrations.

The ionization coefficient as a function of temperature from different models is shown in Fig. 2 at an electric field, $F = 2 \times 10^6$ V/cm, in the temperature range of 300...1500 K. Experimental temperature ranges of the respective experiments are indicated by the shaded regions.

A much stronger effect of temperature on α is given by the state-of-the-art models compared to the proposed model. The difference is due to the calibration at lower temperatures. In the case of Ragunathan and Baliga [11], above approximately 450 K, the impact ionization coefficient for holes results in a much steeper decrease and goes to a negative value above 588 K.

In the case of Hatakeyama [10], the effect of LO phonon scattering is incorporated in coefficients a and b of α . However, the effect of bandgap narrowing on the threshold energy, described in Section II-B, is not included. This results in a much steeper dependence on temperature. The proposed model explicitly includes both effects and, thus, results in a much weaker dependence on temperature.

In Niwa's model, a strong effect is seen in the impact ionization coefficient of holes compared with the proposed model. However, for electrons, the impact ionization coefficient is assumed to be constant as a function of temperature. This assumption was extracted from calibrations below 425 K. Kimoto *et al.* [31] are currently investigating this phenomenon and believe that this is due to conduction band minigaps (that

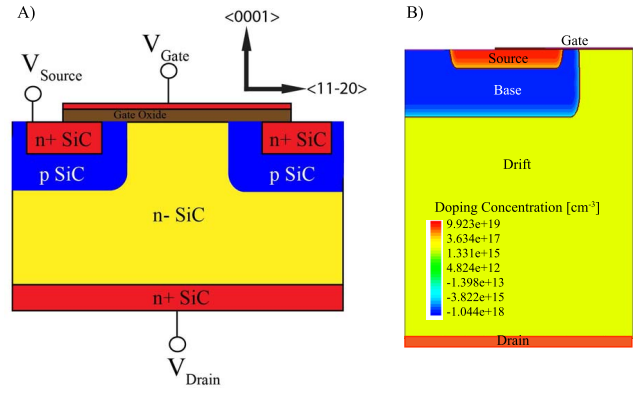


Fig. 3. (a) Full-pitch schematic of the simulated power MOSFET. (b) Power MOSFET half-pitch structure and doping concentration simulated in this paper.

form due to zone-folding) along the (0001) direction shrinking at high temperature.

IV. EFFECTS ON SIMULATION OF DEVICE PERFORMANCE

In this paper, a structure assumed to represent a commercial 1.2-kV, 80-m Ω MOSFET from Wolfspeed is simulated (Fig. 3). The electrical and structural parameters of the MOSFET are extracted to fit the information on the data sheet. The models by Hatakeyama [10] and an Okuto-Crowell fit based on the results of Niwa *et al.* [12] in (0001) direction are implemented in the standard models by Sentaurus TCAD [32]. The proposed model is implemented using the physical model interface available in the software as a user input option. The parameters used for 4H-SiC are from advanced calibrations provided by Sentaurus TCAD unless explicitly stated in the text.

The temperature coefficient of V_{BR} is first investigated with isothermal simulations. A positive temperature coefficient is expected because of the competing phonon scattering mechanisms that reduce the rate of the impact ionization [6], [33]. In this simulation, terminal voltages of the MOSFET are increased until it conducts 20 A of current through the avalanche breakdown. The device temperature, assumed homogeneous, is then increased up to T_{int} and results are demonstrated in Fig. 4(a) for different impact ionization models. The device is assumed to lose its blocking capability ($V_{BR} = 0$) at T_{int} .

The Hatakeyama model greatly overestimates the temperature coefficient of V_{BR} . The assumption of a temperature-independent α_n in Niwa's model gives electron dominated impact ionization above 1100 K. In this case, avalanche becomes dominated by electrons, which can be directly supplied from the source. Thus, the current supplied by an avalanche and the corresponding V_{BR} decreases. In Fig. 4(b), this effect can be seen with V_{BR} decreasing above approximately 1100 K.

Electrothermal simulations are next compared with avalanche experiments performed on this commercial power MOSFET [34]. The avalanche phenomenon is simulated and

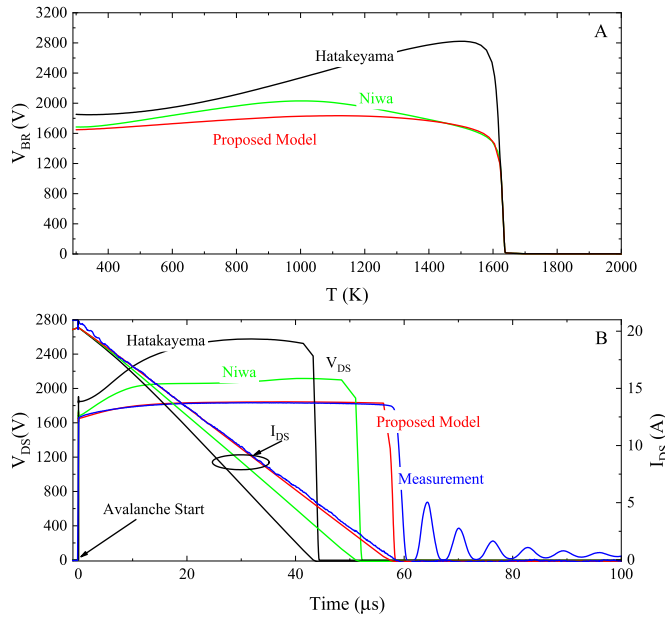


Fig. 4. (a) Evolution of breakdown voltage of a MOSFET required to sustain a current of 20 A as a function of temperature. (b) UIS waveform from measurement as well as simulations using different models.

measured in power MOSFETs using an UIS test as described in [1]. In this circuit, the device under test is used to switch off the current of an inductive load. Switching off inductive loads induces transient EMF and pushes the device into avalanche conduction.

In the second simulation [Fig. 4(b)], an UIS test circuit is simulated where the MOSFET is used to charge a 5.1-mH inductor to 20 A with the gate voltage (V_{GS}) biased at 20 V corresponding to the measurement setup used. The gate is then turned off driving the device into avalanche. Results of the simulations are analyzed to determine the time evolution of the drain current, I_{DS} , and breakdown voltage, V_{BR} , as well as the related transport characteristics during the avalanche event. It is shown that during avalanche, V_{BR} increases from 1.85 kV to a maximum of 2.58 kV ($\Delta V = 730$ V) when using Hatakeyama [10] model, from 1.68 to 2.19 kV ($\Delta V = 510$ V) when using Niwa *et al.* [12], and from 1.65 to 1.84 kV ($\Delta V = 190$ V) when using the proposed model [Fig. 4(b)].

Measurement resulted in an increase from 1.67 to 1.82 kV ($\Delta V = 150$ V) [Fig. 4(b)]. The proposed model provides an excellent match to the UIS measurement.

Therefore, we conclude that the proposed model gives a more accurate representation of avalanche phenomenon at high temperatures. It is the only model that can be applied for avalanche triggered destructive device failure predictions in 4H-SiC.

Understanding the avalanche breakdown mechanism is critical to device design and optimization of ruggedness. Reliability of 4H-SiC devices, such as Schottky-barrier diodes [35] as well as planar and trench power MOSFETs [36] under avalanche conditions benefits from accurate inclusion of anisotropy as well as temperature effects. In addition, analysis of lateral 4H-SiC MOSFETs where transport is mainly

perpendicular to the c -axis may benefit from the inclusion of anisotropy since nominal calibrations are done along the c -axis.

V. CONCLUSION

An improved model for high-temperature avalanche breakdown coefficients in 4H-SiC is presented. Calibration of parameters at room temperature is sufficient to determine the temperature dependence of the avalanche coefficients up to intrinsic temperatures. The anisotropy of impact ionization in 4H-SiC is included in the avalanche model to enable simulations along different orientations from calibrations performed only along one axis. The proposed model, for the first time, gives an excellent fit to measurements of breakdown voltages, which allows predictions of avalanche ruggedness. It is also applicable to avalanche breakdown of other wide bandgap semiconductors, such as GaN, diamond, and Ga_2O_3 , where the critical electric field is high and where under high scattering rate, the collision broadening may become critical.

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REFERENCES

- [1] M. D. Kelley, B. N. Pushpakaran, and S. B. Bayne, "Single-pulse avalanche mode robustness of commercial 1200 V/80 mΩ SiC MOSFETs," *IEEE Trans. Power Electron.*, vol. 32, no. 8, pp. 6405–6415, Aug. 2017.
- [2] S. K. Ghandhi, *Semiconductor Power Devices: Physics of Operation and Fabrication Technology*, 1st ed. Hoboken, NJ, USA: Wiley, 1977.
- [3] B. J. Baliga, *Fundamentals of Power Semiconductor Devices*. New York, NY, USA: Springer, 2008.
- [4] T. Kimoto, "Material science and device physics in SiC technology for high-voltage power devices," *Jpn. J. Appl. Phys.*, vol. 54, pp. 040103-1–040103-27, Apr. 2015.
- [5] M. Bhatnagar and B. J. Baliga, "Comparison of 6H-SiC, 3C-SiC, and Si for power devices," *IEEE Trans. Electron Devices*, vol. 40, no. 3, pp. 645–655, Mar. 1993.
- [6] L. V. Keldysh, "Concerning the theory of impact ionization in semiconductors," *Sov. Phys. JETP*, vol. 21, no. 6, p. 1135, 1965.
- [7] W. Shockley and W. T. Read, Jr., "Statistics of the recombinations of holes and electrons," *Phys. Rev.*, vol. 87, no. 46, pp. 835–842, 1952.
- [8] G. A. Baraff, "Distribution functions and ionization rates for hot electrons in semiconductors," *Phys. Rev.*, vol. 128, no. 6, pp. 2507–2517, 1962.
- [9] P. A. Wolff, "Theory of electron multiplication in silicon and germanium," *Phys. Rev.*, vol. 95, no. 6, pp. 1415–1431, 1954.
- [10] T. Hatakeyama, "Measurements of impact ionization coefficients of electrons and holes in 4H-SiC and their application to device simulation," *Physica Status Solidi A, Appl. Mater. Sci.*, vol. 206, no. 10, pp. 2284–2294, 2009.
- [11] R. Raghunathan and B. Baliga, "Temperature dependence of hole impact ionization coefficients in 4H and 6H-SiC," *Solid-State Electron.*, vol. 43, no. 2, pp. 199–211, Feb. 1999.
- [12] H. Niwa, J. Suda, and T. Kimoto, "Impact ionization coefficients in 4H-SiC toward ultrahigh-voltage power devices," *IEEE Trans. Electron Devices*, vol. 62, no. 10, pp. 3326–3333, Oct. 2015.
- [13] W. S. Loh *et al.*, "Impact ionization coefficients in 4H-SiC," *IEEE Trans. Electron Devices*, vol. 55, no. 8, pp. 1984–1990, Aug. 2008.
- [14] T. Kimoto, *Fundamentals of Silicon Carbide Technology*. Singapore: Wiley, 2014.
- [15] K. K. Thornber, "Applications of scaling to problems in high-field electronic transport," *J. Appl. Phys.*, vol. 52, no. 1, pp. 279–290, 1981.
- [16] A. O. Konstantinov, Q. Wahab, N. Nordell, and U. Lindefelt, "Ionization rates and critical fields in 4H silicon carbide," *Appl. Phys. Lett.*, vol. 71, no. 1, p. 90, 1997.

- [17] A. G. Chynoweth, "Ionization rates for electrons and holes in silicon," *Phys. Rev.*, vol. 109, no. 5, pp. 1537–1540, 1957.
- [18] Y. Okuto and C. R. Crowell, "Threshold energy effect on avalanche breakdown voltage in semiconductor junctions," *Solid-State Electron.*, vol. 18, pp. 161–168, Feb. 1975.
- [19] A. Matulionis, J. Liberis, I. Matulionienė, H.-Y. Cha, L. F. Eastman, and M. G. Spencer, "Hot-phonon temperature and lifetime in biased 4H-SiC," *J. Appl. Phys.*, vol. 96, no. 11, pp. 6439–6444, 2004.
- [20] A. Akturk, N. Goldsman, S. Potbhare, and A. Lelis, "High field density-functional-theory based Monte Carlo: 4H-SiC impact ionization and velocity saturation," *J. Appl. Phys.*, vol. 105, no. 3, 2009, Art. no. 033703.
- [21] P. A. Ivanov, A. S. Potapov, T. P. Samsonova, and I. V. Grekhov, "Electric-field dependence of electron drift velocity in 4H-SiC," *Solid-State Electron.*, vol. 123, pp. 15–18, Sep. 2016.
- [22] I. A. Khan and J. A. Cooper, "Measurement of high-field electron transport in silicon carbide," *IEEE Trans. Electron Devices*, vol. 47, no. 2, pp. 269–273, Feb. 2000.
- [23] F. Capasso, "Physics of avalanche photodiodes," *Semicond. Semimetals*, vol. 22, pp. 1–172, 1985.
- [24] A. Akturk, J. M. McGarrity, S. Potbhare, and N. Goldsman, "Radiation effects in commercial 1200 v 24 A silicon carbide power MOSFETs," *IEEE Trans. Nucl. Sci.*, vol. 59, no. 6, pp. 3258–3264, Dec. 2012.
- [25] K. Hess, *Monte Carlo Device Simulation: Full Band and Beyond*. New York, NY, USA: Springer, 1991.
- [26] C. R. Crowell and S. M. Sze, "Temperature dependence of avalanche multiplication in semiconductors," *Appl. Phys. Lett.*, vol. 9, no. 6, pp. 242–244, 1966.
- [27] Y. Peng, X. Hu, X. Xu, X. Chen, J. Peng, J. Han, and S. Dimitrijević, "Temperature and doping dependence of the Raman scattering in 4H-SiC," *Opt. Mater. Express*, vol. 6, no. 9, pp. 2725–2733, 2016.
- [28] Y. Okuto and C. R. Crowell, "Energy-conservation considerations in the characterization of impact ionization in semiconductors," *Phys. Rev. B, Condens. Matter*, vol. 6, no. 8, pp. 3076–3081, 1972.
- [29] M. E. Levinstein, S. L. Rumyantsev, and M. Shur, *Properties of Advanced Semiconductor Materials: GaN, AlN, InN, BN, SiC, SiGe*. Hoboken, NJ, USA: Wiley, 2001.
- [30] W. J. Schaffer, G. H. Negley, K. G. Irvine, and J. W. Palmour, "Conductivity anisotropy in epitaxial 6H and 4H SiC," *Mater. Res.*, vol. 339, no. 1, pp. 595–600, 2017.
- [31] T. Kimoto, H. Niwa, T. Okuda, E. Saito, Y. Zhao, S. Asada, and J. Suda, "Carrier lifetime and breakdown phenomena in SiC power device material," *J. Phys. D: Appl. Phys.*, vol. 51, no. 36, 2018, Art. no. 363001.
- [32] *Advanced Calibration for Device Simulation*, Synopsys, Mountain View, CA, USA, Sep. 2017.
- [33] C. L. Anderson and C. R. Crowell, "Threshold energies for electron-hole pair production by impact ionization in semiconductors," *Phys. Rev. B, Condens. Matter*, vol. 5, no. 6, pp. 2267–2272, 1972.
- [34] *C2M0080120D Silicon Carbide Power MOSFET C2M MOSFET Technology, Datasheet*, CREE, Durham, NC, USA, 2015, pp. 1–10.
- [35] A. Konstantinov *et al.*, "Investigation of avalanche ruggedness of 650 V Schottky-barrier rectifiers," *Solid-State Electron.*, vol. 148, pp. 51–57, Oct. 2018.
- [36] D. Peters *et al.*, "The new CoolSiCTM trench MOSFET technology for low gate oxide stress and high performance," in *Proc. PCIM Eur.*, 2017, pp. 1–7.